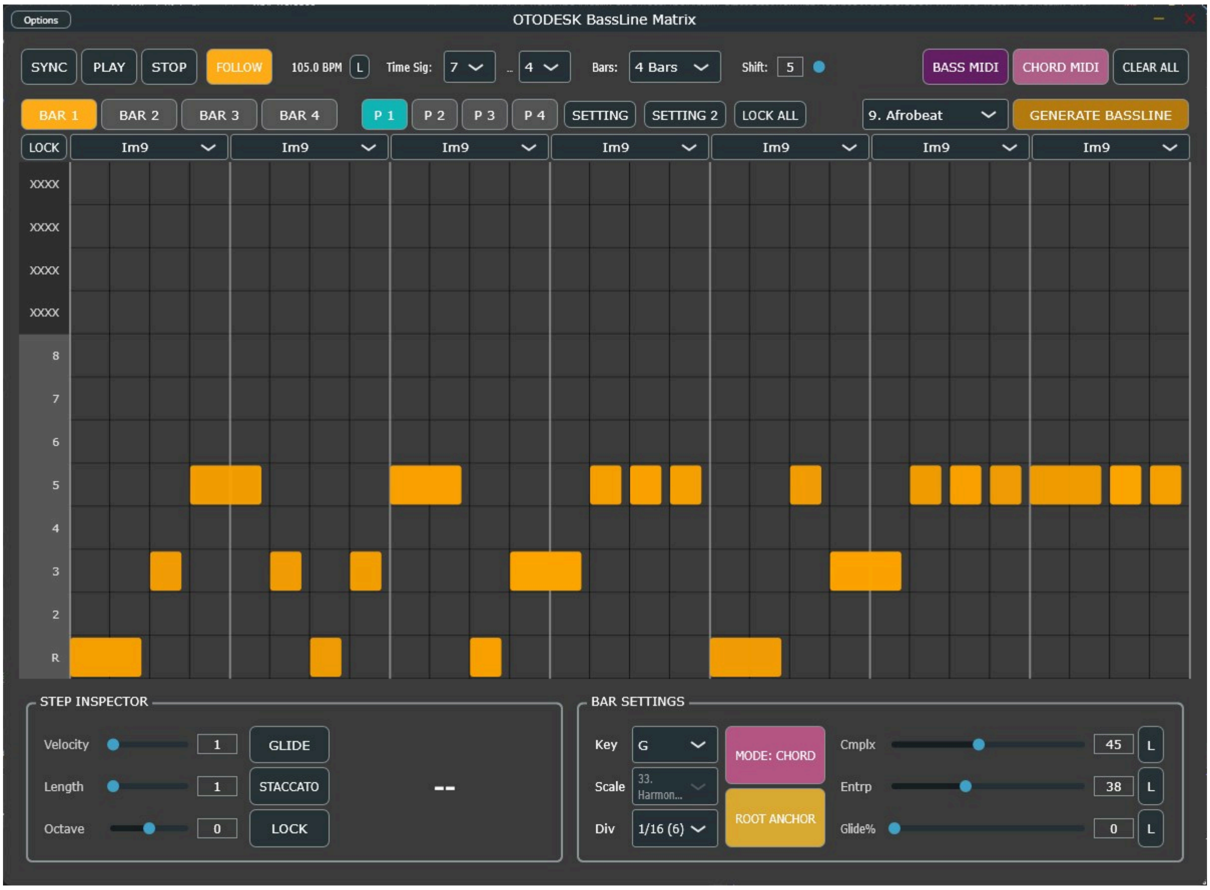

BassLineMatrix - Official User Manual

Welcome to **BassLineMatrix**, an advanced algorithmic MIDI sequencer and built-in synthesizer plugin designed for generating complex, genre-specific basslines and chord progressions across 23 distinct musical styles. Driven by a highly optimized, real-time safe C++ architecture, BassLineMatrix serves as both an ultimate ideation tool and a reliable live-performance instrument.

This manual covers all parameters, operational guidelines, and the sound engine's detailed settings to help you maximize the plugin's potential.

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1. Top Bar & Global Controls

The top section of the interface governs the global playback, timing, and structural settings of your pattern.

- **SYNC**: Synchronizes the internal sequencer to your DAW's playhead and tempo. When active, playback is completely driven by the host DAW.

- **PLAY / STOP**: Manual transport controls for the internal sequencer (useful when working standalone or with SYNC disabled).
- **FOLLOW**: When enabled, the UI will automatically switch to the currently playing Bar tab, keeping the active grid in view.
- **Tempo & Lock (L)**: Sets the internal BPM. Click the **L** (Lock) button to prevent the tempo from changing when you select a new genre. (Displays "DAW BPM" when SYNC is on).
- **Time Sig**: Sets the global time signature (e.g., 4/4, 5/8). The sequencer matrix dynamically adjusts its tick resolution based on the denominator.
- **Bars**: Sets the total length of the sequence loop (1 to 4 Bars).
- **Shift**: A global pitch shifter (-24 to +24 semitones) applied to both the Bass and Chord MIDI outputs.
- **CLEAR ALL**: Erases all unlocked notes across the current slot.
- **P 1 / P 2 / P 3 / P 4 (Pattern Slots)**: Four independent memory slots to store and switch between different pattern variations instantly.
- **LOCK ALL**: A global master switch to lock (protect from regeneration) or unlock all currently active notes in the matrix for the selected slot.

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2. Sequencer Grid & Main Editor

The central matrix is where your basslines come to life. The Y-axis represents pitch (mapped to the selected scale or chord), and the X-axis represents time (ticks).

Mouse & Keyboard Operations

- **Click & Drag**: Click an empty space to add a note. Drag horizontally to adjust its length.
- **Right-Click**: Toggles the **Staccato** property of a note.
- **Shift + Click**: Toggles the **Glide** property of a note.
- **Mouse Wheel**: Scroll over a note to adjust its **Octave**. Hold `Ctrl` (or `Cmd`) and scroll to adjust its **Velocity** (opacity will change).
- **Arrow Keys**: Navigate the selected note around the grid.
- **L Key**: Instantly locks/unlocks the currently selected note.

Visual Feedback

- **Opacity**: Indicates Note Velocity (brighter = harder).
- **Width**: Indicates Note Length.
- **Cyan Color**: Indicates Glide is active.
- **Orange Color**: Standard note playback.
- **Red Border**: The note is Locked and will not be overwritten during generation.
- **White Border**: Currently selected note.
- **Text (+/- / .)**: Displays octave offsets (e.g., +1, -1) and a dot (`.`) if Staccato is active.

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3. Step Inspector

Located at the bottom left, the Step Inspector provides precise control over the currently selected note in the grid.

- **Velocity [1 - 127]**: The MIDI velocity of the note.
- **Length [1 - 96]**: The duration of the note in sequencer ticks.
- **Octave [-2 to +2]**: Shifts the individual note up or down by octaves.
- **GLIDE Toggle**: Enables a portamento slide from the previous note.
- **STACCATO Toggle**: Shortens the actual playback gate time based on the global Staccato Ratio.
- **LOCK Toggle**: Protects the note from being erased or altered when clicking "GENERATE BASSLINE".
- **Note Name Display**: Shows the absolute MIDI pitch of the selected note (e.g., C3, F#2), factoring in the key, scale, octave, and global shift.

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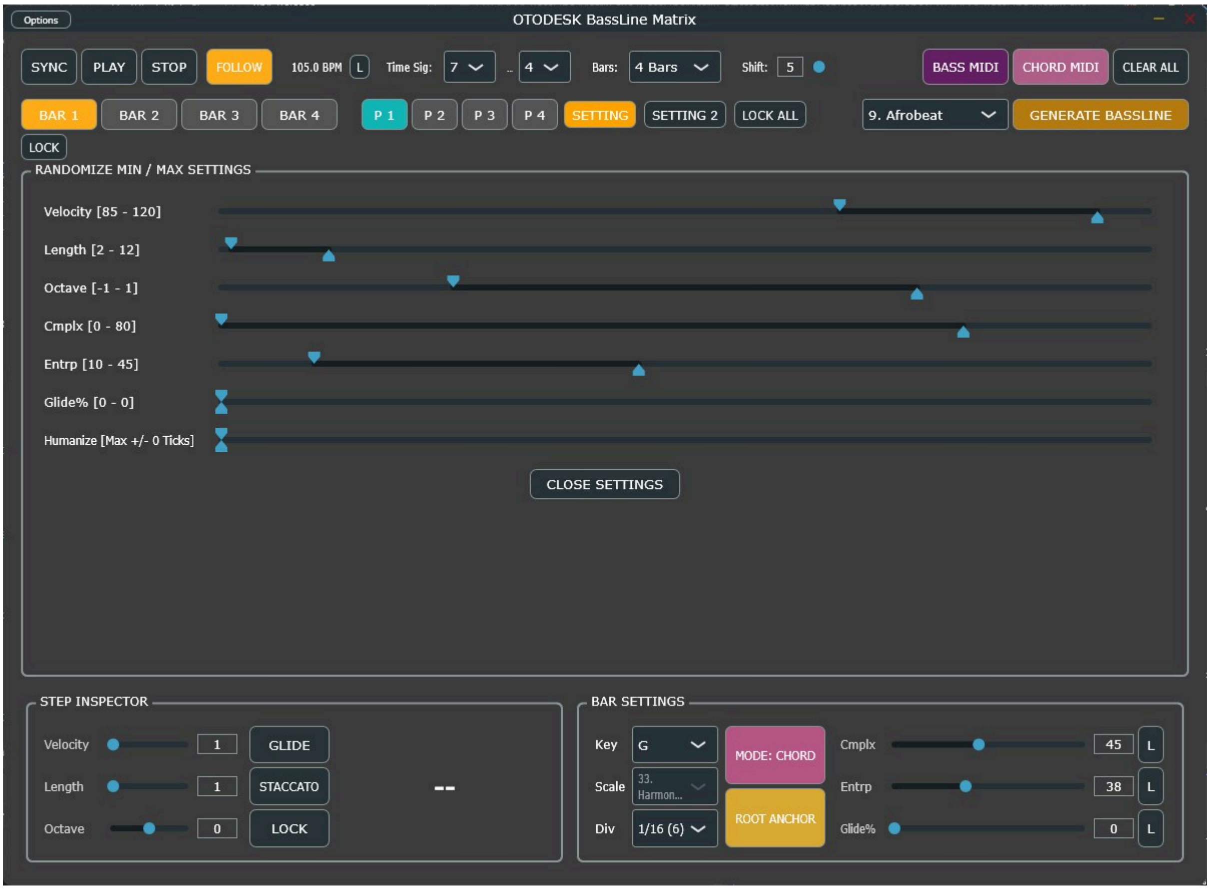
4. Bar Settings & Generation Controls

These parameters define the musical context and behavioral probabilities for the procedural generation engine. They are set *per bar*.

- **Key**: The root note of the scale/chord (C to B).
- **Scale / Chord Menu**: Depending on the active Mode, selects the musical scale (40 variations) or the Chord Quality (15 variations including Maj9, Dom13, HalfDim, etc.).
- **Div**: The grid resolution and quantization snap value (e.g., 1/8, 1/16, 1/32, 1/16T, 1/48, 1/96).
- **MODE: SCALE / CHORD**: Toggles how the Y-axis of the grid is interpreted.
  - *SCALE*: The grid maps to the degrees of the selected scale.
  - *CHORD*: The grid maps to specific chord tones and extensions, adapting perfectly to the harmonic progression.
- **LOCK (Chord)**: Prevents the chord progression from being overwritten during generation.
- **Cmplx (Complexity)**: Determines the density and rhythmic busyness of the generated pattern.
- **Entrp (Entropy)**: Determines the unpredictability, syncopation, and probability of notes jumping to different octaves or playing "out-of-scale" passing tones.
- **Glide%**: The probability (0-100%) that a generated note will have Glide enabled.
- *(Note: The `L` buttons next to Cmplx, Entrp, and Glide% lock these sliders from being randomized during generation).*
- **ROOT ANCHOR**: When enabled, the generator ensures a strong root note is placed at the downbeat of the bar to maintain musical stability.
- **GENERATE BASSLINE**: Triggers the algorithmic DNA engine to populate the grid based on the selected Genre, Complexity, and Entropy settings.
- **Genre Menu**: Selects the stylistic DNA for the generator (23 genres including Techno, Dubstep, Neo Soul, Amapiano, etc.).

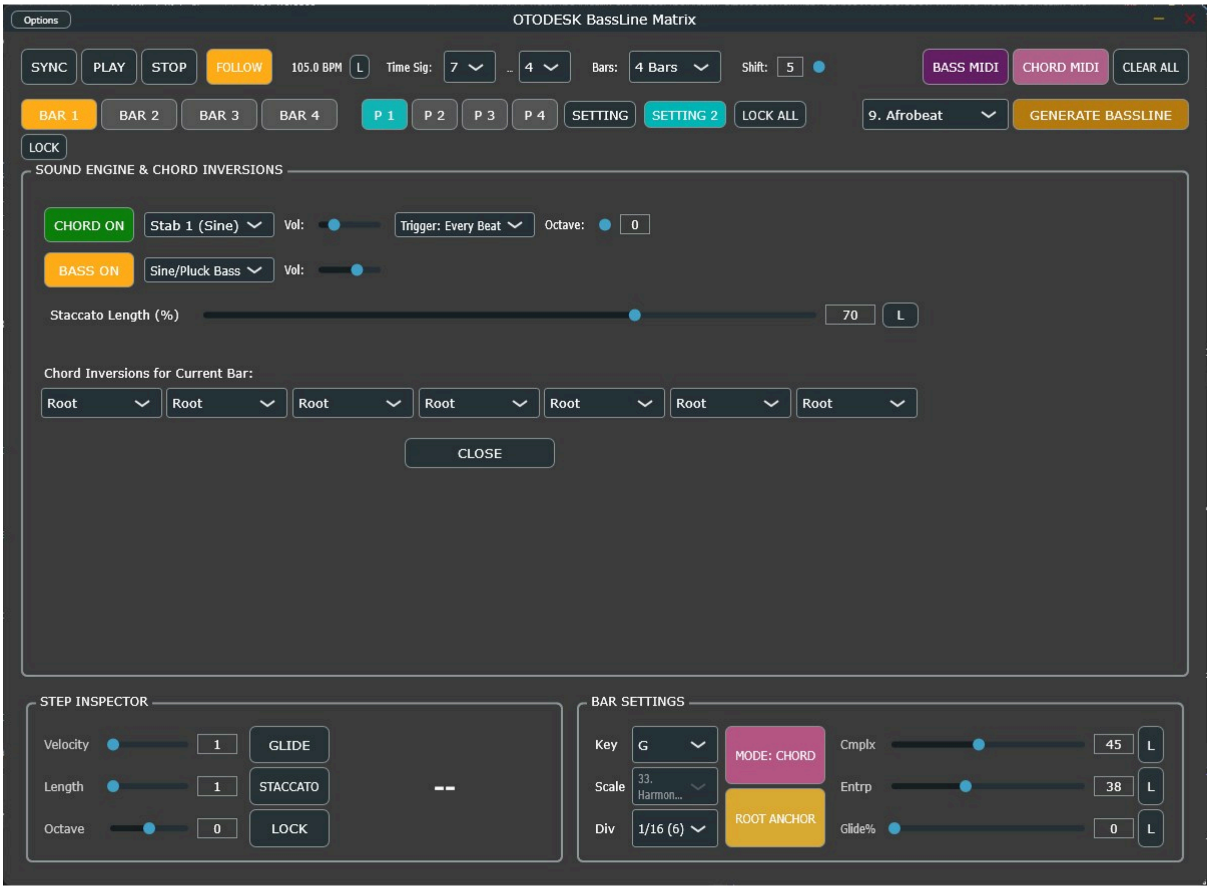
5. Setting 1: Randomize Min / Max Settings



Accessed via the **SETTING** button, this panel defines the global boundaries for the procedural generation engine. When "GENERATE BASSLINE" is clicked, the engine will pick values within these ranges.

- **Velocity [Min - Max]**: The dynamic range for generated notes.
- **Length [Min - Max]**: The minimum and maximum tick length for generated notes.
- **Octave [Min - Max]**: The allowed octave displacement (-2 to +2).
- **Cmplx [Min - Max]**: The range from which the per-bar Complexity slider will be randomized.
- **Entrp [Min - Max]**: The range for the Entropy randomization.
- **Glide% [Min - Max]**: The range for the Auto-Glide probability.
- **Humanize [Max +/- Ticks]**: Injects microscopic timing variations (jitter) to the note offsets, shifting them slightly off the perfect grid. Essential for Lo-Fi, Neo Soul, and Groove-heavy genres.

6. Setting 2: Sound Engine & Chord Inversions



Accessed via the **SETTING 2** button, this panel controls the internal Zero-Delay Feedback (ZDF) / Topology-Preserving Transform (TPT) synthesizer engine and harmonic structures.

Internal Synth Controls

- **CHORD ON / BASS ON**: Toggles the internal synthesizer voices on or off. Turn these off if you are only using BassLineMatrix as a MIDI sequencer for external plugins.
- **Sound Menus**: Select the oscillator waveforms.
  - *Bass*: Sine/Pluck, Saw, Square Sub, Acoustic.
  - *Chord*: Stab 1-3, Pad 1-3.
- **Vol**: Independent volume control for the Bass and Chord synth engines.
- **Chord Trigger Mode**:
  - *On Change*: Chords only re-trigger when the chord changes in the progression.
  - *Every Beat*: Chords trigger on every downbeat.
- **Chord Octave**: Shifts the entire chord progression up or down by octaves.
- **Staccato Length (%)**: Determines the actual gate time of notes that have the Staccato property enabled (e.g., 30% of the visual note length).
- **L (Staccato Lock)**: Prevents the Staccato Length from changing when selecting a new Genre.

Harmonic Controls

- **Chord Inversions**: A row of dropdown menus corresponding to each beat of the current bar. Allows you to set the inversion (Root, 1st, 2nd, 3rd) of the chord, enabling smooth voice leading.
- **Recommended Scales**: Displays a helpful list of suggested scales perfectly suited for the currently selected Genre.

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7. MIDI Export

BassLineMatrix is fundamentally a MIDI generation powerhouse. Once you are satisfied with your pattern, you can route the MIDI out of the plugin directly, or use the drag-and-drop features.

- **BASS MIDI**: Click and drag this button into your DAW's timeline to export the monophonic bass sequence as a standard `.mid` file. Humanize offsets, staccato gate times, and glide overlaps are strictly calculated and baked into the exported file.
- **CHORD MIDI**: Click and drag this button to export the polyphonic chord progression (up to 5-voice extensions based on your Chord Mode settings) into your DAW.

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8. Disclaimer & Real-Time Safety

BassLineMatrix is built on a highly optimized, lock-free double-buffering architecture designed to ensure zero audio dropouts, even when regenerating complex patterns during live playback.

However, because the internal synthesizer utilizes mathematically pure, non-bandlimited oscillators and zero-delay feedback (ZDF) filters to maintain zero latency and preserve extreme transient punch, extreme parameter combinations may result in high harmonic content. Always employ standard gain staging practices and use a limiter on your master bus when experimenting with high velocities and rapid sequence generation.